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MedBooking: An Online Medical Appointment Booking System Integrating AI Chatbot and Telemedicine

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Abstract. The increasing demand for efficient healthcare services has exposed critical shortcomings in traditional appointment systems: fragmented scheduling across multiple providers, long wait times, and the absence of real-time availability information. This paper presents **MedBooking**, a web-based medical appointment booking platform developed at FPT University as part of the SWP391 capstone project. The system provides patients with centralised scheduling, real-time slot visibility, AI-powered symptom triage, integrated telemedicine, and automated multi-channel reminders with one-click cancellation. Built on a React.js frontend, a Node.js/Express.js backend, and a MongoDB database, MedBooking is the only platform among its Vietnamese counterparts to combine booking, telemedicine, AI chatbot, e-pharmacy, OCR prescription import, and an interactive medical map in a single application. A review of peer-reviewed papers from Springer confirms that all major design decisions are grounded in empirically validated findings on digital health adoption and patient-centred scheduling.

Keywords: Medical Appointment Booking · Telemedicine · AI Chatbot · Healthcare Information System · Digital Health · Patient-Centred Care

1 Introduction

Healthcare accessibility remains a global challenge. In Vietnam, patients frequently navigate fragmented booking channels — calling multiple clinics, waiting for manual confirmations, and receiving no automated follow-up — resulting in high no-show rates and poor user satisfaction. A study on MRI scheduling found that two-thirds of patients waited more than one month for an appointment, with the absence of a centralised booking point as the primary barrier [8].

Digital health platforms have emerged as a promising solution. Yet research consistently shows that perceived usefulness and ease of use are the two strongest predictors of adoption, particularly among older adults [1]. Systems that are technically capable but difficult to use fail in practice [5].

Competitive gap. Table 3 (Section 3) shows that existing Vietnamese platforms each cover at most three of six key feature areas. BookingCare offers booking and a medical map but lacks telemedicine and AI; eDoctor adds telemedicine and e-pharmacy but has no chatbot or map. No single platform combines all six capabilities.

Contribution. MedBooking addresses this gap by delivering:

1. A four-step centralised booking flow with real-time slot availability (FR-10 to FR-13).
2. An AI chatbot for symptom-based specialty recommendation (FR-33).
3. Integrated telemedicine via WebRTC video call and Socket.io chat (FR-30, FR-31).
4. Evidence-based automated reminders with one-click cancel/reschedule links (FR-16).
5. OCR prescription import and e-pharmacy integration (FR-32, FR-34).
6. A Google Maps medical facility locator (FR-35).

The remainder of this paper is organised as follows. Section 2 reviews related work. Section 3 describes the system architecture and features. Section 4 presents the system design methodology. Section 5 discusses alignment between the literature and our design. Section 6 concludes with future directions.

2 Related Work

2.1 Telemedicine Adoption and the TAM Model

He et al. [1] surveyed older adults in China using the Technology Acceptance Model (TAM) combined with emotional design theory, analysed via structural equation modelling. *Perceived usefulness* and *perceived ease of use* were the strongest adoption predictors; technological anxiety had a significant negative effect [1]. These results informed MedBooking's UX priorities: a four-step booking flow, minimum font size, and clear contrast.

Rafi [5] conducted a mixed-methods study with elderly participants. Although a significant portion reported using at least one AI tool, low digital literacy and lack of trust remained major barriers [5]. The study directly motivated MedBooking's proxy booking feature, which allows users to manage appointments on behalf of less tech-savvy relatives.

2.2 Centralised Scheduling and Wait-Time Reduction

Singer et al. [8] analysed scheduling experiences, noting that long wait times delay diagnosis and treatment. The root cause was often the absence of a unified booking point, leading to a strong recommendation for a centralised contact hub to streamline the process [8]. This constitutes the core architectural principle of MedBooking.

Kaluza et al. [2] modelled online booking systems within a multinomial logit framework. A key finding emerged: exposing *all* available slots can counterintuitively reduce bookings because users infer low demand and lower quality from too many empty slots [2]. Their recommendation to expose only a controlled subset of slots directly shapes MedBooking's slot-display strategy.

2.3 Evidence-Based Reminder Design

Selim et al. [3] ran an experiment evaluating consumer preferences for healthcare appointment reminders. The analysis identified timely delivery before the appointment and a direct cancellation/reschedule link as critical components to increase acceptance and reduce no-shows [3]. MedBooking implements exactly these attributes.

2.4 AI, NLP, and Clinical Decision Support

Sharma [6] reviewed AI, robotics, and NLP applications advancing telemedicine and remote patient monitoring. The study demonstrates that NLP enables efficient symptom triage, personalised recommendations, and virtual consultation support [6]. This constitutes the theoretical foundation for MedBooking's AI chatbot module.

Silva-Ferreira et al. [7] conducted a scoping review of AI studies in healthcare contexts. A key finding is that ethical, legal, and data-privacy barriers — not just technical limitations — are the primary obstacles to adoption [7]. MedBooking addresses these concerns through encrypted authentication, password hashing, and secure token storage.

2.5 Stakeholder Requirements and System Quality

Ostadmohammadi et al. [4] analysed perceptions of electronic appointment systems across multiple stakeholder groups using qualitative content analysis. Problems fell into both management and non-management categories, showing that overall satisfaction depends heavily on both functional performance and user experiences [4]. MedBooking addresses this by balancing comprehensive features with strict non-functional requirements.

3 System Architecture and Key Features

3.1 Three-Tier Architecture

MedBooking follows a standard three-tier web architecture:

- **Presentation layer** — React.js single-page application, responsive for desktop and mobile browsers.
- **Application layer** — Node.js/Express.js RESTful API server handling business logic, WebRTC signalling via Socket.io, and third-party integrations (Google Maps, Tesseract.js OCR, email/SMS notification services).
- **Data layer** — MongoDB document store for flexible, schema-less medical records; MongoDB Atlas for cloud hosting.

3.2 Actor Model

Table 1 lists the primary actors based on stakeholder roles identified in the literature.

Table 1: System actors and literature basis

Actor	Role	Ref.
Patient	Books appointments; uses chatbot, telemedicine, health records	[8,3,4]
Doctor	Sets schedule; manages consultations; fills examination forms	[4]
Admin	Approves accounts; manages specialties; views statistics	[4]
Clinic Manager	Registers facility; manages affiliated doctors	[4]
Email System	External actor; executes automated notification commands	[3]

3.3 Core Functional Requirements

Table 2 presents the primary functional requirements, each grounded in the literature review.

Table 2: Core functional requirements with research basis

ID	Feature	Description	Ref.
FR-01	Registration	Email + password sign-up with OTP verification	—
FR-05	Relative mgmt.	Proxy booking for family members (elderly support)	[5]
FR-10	Doctor search	Filter by specialty, name, hospital, GPS location	[8]
FR-11	Slot display	Real-time slot visibility with controlled subset	[2]
FR-16	Auto reminder	Email + SMS notifications; one-click cancel/reschedule	[3]
FR-30	Video call	WebRTC real-time video consultation	[1,6]
FR-32	OCR prescription	Tesseract.js-based prescription import from image/PDF	[6]
FR-33	AI Chatbot	Symptom triage → specialty recommendation	[6,7]
FR-34	E-pharmacy	In-app pharmacy ordering linked to prescription	[6]
FR-35	Medical map	Google Maps locator for nearby healthcare facilities	[8]

3.4 Competitive Analysis

Table 3 compares MedBooking with competing Vietnamese platforms across six key capability areas.

Table 3: Feature comparison with competing Vietnamese platforms

Platform	Booking	Telemd.	AI Bot	E-Pharm.	Map	OCR
BookingCare	✓	–	–	–	✓	–
eDoctor	✓	✓	–	✓	–	–
MedBooking	✓	✓	✓	✓	✓	✓

4 Method

4.1 Research-by-Learning (RBL) Approach

MedBooking was developed following a *Research-by-Learning* (RBL) methodology [1,4], in which each design decision was grounded in a peer-reviewed finding before implementation began. The process comprised four phases:

1. **Literature review** — Ten peer-reviewed papers were collected from Springer and analysed for findings related to telemedicine adoption, scheduling optimisation, reminder design, and AI in healthcare.
2. **Requirements elicitation** — Insights from the literature were translated into functional and non-functional requirements documented in the Software Requirements Specification (SRS).
3. **System design** — The SRS informed the Entity Relationship Diagram (ERD), Use Case model, and UI screen flow, all of which were iteratively refined.
4. **Implementation and validation** — The system was built incrementally using an Agile-inspired sprint structure and tracked on Jira; artefacts were committed to a shared Git repository.

4.2 System Design Artefacts

Entity Relationship Diagram (ERD). The database schema captures ten core entities: `User`, `Doctor`, `Patient`, `Clinic`, `Specialty`, `Appointment`, `TimeSlot`, `Prescription`, `LabTest`, and `Notification`. Key relationships include:

- A `Doctor` belongs to one `Clinic` and one `Specialty`.
- An `Appointment` links one `Patient` to one `Doctor` via one `TimeSlot`.
- A `Prescription` is issued per `Appointment` and may trigger an e-pharmacy order.

Use Case Model. The system defines 30 use cases across four actor groups (Table 1). The most critical flows are:

- **UC-P18 Book Appointment** — Patient selects specialty → searches doctor → selects slot → confirms booking → receives email confirmation.
- **UC-D12 Fill Examination Form** — Doctor records vital signs and symptoms, optionally orders lab tests (UC-D13), then issues diagnosis and prescription.
- **UC-P15 AI Specialty Suggestion** — System analyses patient onboarding survey and visit history to recommend the most relevant specialty before booking.

Technology Stack. Table 4 summarises the implementation technology choices and the rationale for each.

Proposed System Architecture. The proposed MedBooking system is built upon a scalable, decoupled architecture to ensure high availability and maintainability. The architecture is divided into three primary functional layers:

- **Client Layer (Frontend):** Developed using React.js, it provides a responsive interface for both patients and doctors. Redux is utilised for global state management (e.g., managing authentication tokens and booking states).
- **Application Layer (Backend):** A Node.js/Express.js RESTful API handles core business logic. It integrates several advanced modules: a Socket.io server for real-time slot updates and WebRTC signalling (telemedicine), an NLP-based engine for the AI Chatbot (FR-33), and third-party gateways for Mail/SMS reminders (FR-16).
- **Data Layer:** MongoDB Atlas serves as the primary database, accommodating heterogeneous medical records (prescriptions, lab tests) via its flexible document schema. Security is enforced using bcrypt for password hashing and JWT for API route protection.

Screens Flow. The user interface flow is designed to minimise cognitive load, directly applying the perceived ease-of-use principles from the TAM model [1]. The core patient navigation flow proceeds as follows:

1. **Landing Page:** Displays system overview, the AI Chatbot widget for symptom triage, and login prompts.
2. **Authentication:** Secure OTP-based login or registration.
3. **Search & Discovery:** Patients locate facilities via the integrated Google Maps locator (FR-35) or filter doctors by the specialty recommended by the AI.
4. **Booking Execution:** Select doctor → view real-time availability (FR-11) → confirm appointment details.
5. **Patient Dashboard:** Post-booking workspace where users can view electronic health records (EHR), access telemedicine video call links (FR-30), or perform one-click cancellations.

Fig. 1: Core screens flow of the MedBooking system for patient appointment booking.

4.3 Evaluation Plan and Expected Outcomes

To ensure the system satisfies both the literature-driven requirements and stakeholder expectations, we propose a multi-faceted evaluation plan:

- **Usability and Adoption (UX):** We will administer the System Usability Scale (SUS) to a cohort of 30 test users across different age groups. *Expected outcome:* A SUS score exceeding 75, validating the design's ease of use for elderly patients.
- **AI Chatbot Accuracy:** The NLP symptom triage model will be tested against 100 historical clinical case descriptions. *Expected outcome:* At least an 85% accuracy rate in mapping user symptoms to the correct medical specialty.
- **System Performance & Concurrency:** Apache JMeter will simulate high-traffic scenarios to test the centralised booking engine. *Expected outcome:* The system maintains a server response time of under 2 seconds for up to 500 concurrent booking requests without data race conditions on time slots.
- **Functional Reliability:** End-to-end testing of the WebRTC video calls and automated reminder delivery. *Expected outcome:* 95% successful connection rate for telemedicine sessions and 100% successful execution of cron jobs for email/SMS notifications.

4.4 Development Process

The project followed a four-week sprint structure aligned with the SWP391 course timeline:

- **Week 1** — Project initialisation: SRS template, README, Git repository setup, Jira task board.

- **Week 2** — Literature review: collected 10 papers from Springer; produced a consolidated analysis document; completed SRS Introduction, Scope, and existing-systems survey.
- **Week 3** — System design: frontend and backend repositories created; database schema implemented in MongoDB; SRS updated with ERD, Use Case diagrams, and UI wireframes; LaTeX template initialised on Overleaf.
- **Week 4** — Paper writing: adopted Springer Scopus Q4 template; drafted Introduction, Related Work, Method, and References sections; ERD and Use Case model refined.

5 Discussion

5.1 Literature–Design Alignment

Every major design decision in MedBooking traces directly to at least one peer-reviewed finding:

- The **four-step UX flow and accessibility guidelines** (minimum font, high contrast) respond to TAM findings on perceived ease of use [1].
- The **centralised booking hub** addresses the unified scheduling point recommended to reduce wait times [8].
- The **controlled slot display** (subset rather than all slots) directly implements the quality-signal model of Kaluza et al. [2].
- The **timed reminder with one-click cancel link** replicates the highest-preference reminder format identified experimentally [3].
- The **NLP-based AI chatbot** applies findings on symptom triage and personalised recommendations [6].
- The **privacy-first architecture** (encrypted auth, hashed passwords, secure tokens) responds to the ethical/legal barriers identified as the primary adoption obstacle [7].
- The **proxy booking feature** for elderly relatives addresses the low digital-literacy barrier reported in mixed-methods research [5].

5.2 Limitations

Several limitations should be acknowledged. First, the system has not yet undergone comprehensive user testing with real patients and doctors; adoption metrics remain projected from the literature [1,4]. Second, the AI chatbot's specialty classification accuracy has not been empirically validated in the Vietnamese medical context. Third, load testing for high concurrent usage is planned but not yet executed. Finally, the OCR prescription pipeline relies on Tesseract.js, which may underperform on handwritten notes.

6 Conclusion

This paper presented MedBooking, a patient-centred online medical appointment booking system developed for the SWP391 capstone course at FPT University, Danang. The design is closely informed by a structured review of peer-reviewed papers spanning telemedicine adoption, scheduling optimisation, reminder systems, and AI implementation in healthcare. MedBooking is unique among its Vietnamese counterparts in combining all six capability areas — booking, telemedicine, AI chatbot, e-pharmacy, OCR, and medical map — within a single platform.

Future work will focus on:

- Usability testing with real clinics and patient cohorts.
- Benchmarking the AI specialty classifier against active clinical datasets in the Vietnamese context.
- Load and stress testing under concurrent booking scenarios.
- Improving OCR accuracy for handwritten Vietnamese prescriptions.

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Table 4: Technology stack and design rationale

Layer	Technology	Rationale
Frontend	React.js	Component-based UI enables fast iteration; supports minimum font size and contrast guidelines from TAM research [1].
Backend	Node.js / Express.js	Non-blocking I/O handles concurrent booking requests; Socket.io enables real-time slot updates and WebRTC signalling [6].
Database	MongoDB Atlas	Document model accommodates heterogeneous medical record schemas without costly migrations.
AI Chatbot	NLP pipeline (keyword + intent classification)	Lightweight model suitable for Vietnamese symptom vocabulary; aligns with NLP findings in literature [6].
OCR	Tesseract.js	Client-side OCR avoids uploading raw prescription images to third-party servers, preserving privacy [7].
Notification	NodeMailer + SMS gateway	Delivers timed multi-channel reminders with one-click cancel link as recommended by [3].